

Primary sources & further reading

Part 11 —  The LLM Model Landscape · 11-llm-models/papers/PAPERS.md · 11-llm-models/resources/RESOURCES.md

Papers & Primary Sources — The LLM Model Landscape

This module leans on benchmark and evaluation methodology papers more than architecture papers (those live in Module 04). Read these to understand what a leaderboard number actually measures.

Benchmark methodology and its limits

Paper	Year	Why it matters	Link
Measuring Massive Multitask Language Understanding (MMLU) — Hendrycks et al.	2020	The benchmark referenced constantly; also read its known-error follow-ups before trusting a score	arXiv:2009.03300
Are We Done With MMLU? — Gema et al.	2024	Documents label errors and near-saturation in MMLU — the "distrust a single score" case, with evidence	arXiv:2406.04127
GPQA: A Graduate-Level Google-Proof Q&A Benchmark — Rein et al.	2023	The harder benchmark that replaced MMLU as a frontier-model differentiator	arXiv:2311.12022
SWE-bench: Can Language Models Resolve Real-World GitHub Issues? — Jimenez et al.	2023	The current standard for coding-agent capability; read the contamination-risk discussion	arXiv:2310.06770
Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference — Chiang et al.	2024	The methodology behind Elo-style leaderboards, and its own documented biases (style, length)	arXiv:2403.04132
HELM: Holistic Evaluation of Language Models — Liang et al.	2022	A multi-metric alternative to single-number leaderboards — worth knowing this exists	arXiv:2211.09110

Contamination and reproducibility

Paper	Year	Why it matters	Link
Data Contamination Quiz: A Tool to Detect and Estimate Contamination in LLMs — Golchin & Surdeanu	2023	A practical method for the exact concern <code>benchmark_literacy_demo.py</code> teaches	arXiv:2311.06233
Time Travel in LLMs: Tracing Data Contamination in Large Language Models — Golchin & Surdeanu	2023	Companion paper, broader detection framework	arXiv:2308.08493

Scaling and the tier structure (deeper treatment in Module 04)

Paper	Year	Link
Training Compute-Optimal Large Language Models (Chinchilla) — Hoffmann et al.	2022	arXiv:2203.15556
Scaling Laws for Neural Language Models — Kaplan et al.	2020	arXiv:2001.08361

Open-weights model families (primary technical reports)

Report	Link
Llama 3 Herd of Models — Meta AI	arXiv:2407.21783
Mistral 7B — Jiang et al.	arXiv:2310.06825
DeepSeek-V3 Technical Report	arXiv:2412.19437
Qwen2.5 Technical Report	arXiv:2412.15115

Model technical reports go stale fast by design — read them for methodology (how they trained, how they evaluated) rather than for the specific numbers, which the next release will supersede.

Resources — The LLM Model Landscape

Live leaderboards and comparisons (bookmark, don't screenshot)

Resource	What it shows	Link
LMSYS Chatbot Arena	Human-preference Elo ratings across providers	https://lmarena.ai/
Artificial Analysis	Independent price/latency/quality comparison across providers — the closest thing to a neutral cross-provider view	https://artificialanalysis.ai/
Hugging Face Open LLM Leaderboard	Open-weights model benchmarks in one place	https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard
Vellum LLM Leaderboard	Task-specific comparisons (coding, reasoning, etc.)	https://www.vellum.ai/llm-leaderboard
SWE-bench leaderboard	Coding-agent capability specifically	https://www.swebench.com/

Official pricing and model pages (the only sources you should treat as authoritative)

Provider	Link
Anthropic	https://claude.com/pricing · https://docs.claude.com/
OpenAI	https://openai.com/api/pricing/ · https://platform.openai.com/docs/models
Google	https://ai.google.dev/pricing
Meta Llama	https://ai.meta.com/llama/
Mistral	https://docs.mistral.ai/
DeepSeek	https://api-docs.deepseek.com/quick_start/pricing/

Courses & deeper reading

Resource	Why	Link
Stanford CS324 — Advances in Foundation Models	Academic treatment of the model landscape and evaluation	https://stanford-cs324.github.io/winter2023/
Hugging Face — Open LLM Leaderboard methodology docs	How to actually read a leaderboard's methodology before trusting it	https://huggingface.co/docs/leaderboards/
"You Don't Need a Bigger Boat"-style engineering blogs on model selection	Search for recent posts from teams documenting a real model-migration decision — more useful than any static list	—

Communities

- [r/LocalLLaMA](#) — active open-weights model tracking and real-world benchmarking
- Hugging Face forums and Discord — model release discussion as it happens
- Provider status pages and changelogs — subscribe directly if a model is load-bearing for your product

Where to go next

- [../04-llm/](#) for how these models are actually built and the theory behind the tiers.
- [../15-ai-evals/](#) for how to build the eval set that should decide your model choice — not a leaderboard.
- [../10-ai-architecture/](#) for the model-gateway pattern that lets you avoid ever betting on just one model.